Spectral Frequency-Unit Utilities Specification

Canonical THz Axis and Derived Angular-Frequency, Wavenumber, and Energy Coordinates

mdstats

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1 Purpose

This document specifies the private module

`mdstats/analysis/_spectral_units.py`

The module converts the canonical spectral frequency axis into equivalent coordinates used in molecular-dynamics and spectroscopy plots. It performs unit conversion only; it never interpolates, resamples, or renormalizes a spectrum.

2 Canonical convention

The canonical frequency is cycles per picosecond:

$$f\ [\mathrm{ps}^{-1}].$$

Numerically,

$$1\ \mathrm{ps}^{-1} = 1\ \mathrm{THz}.$$

All derived axes refer to the same bins.

3 Public status

The module is private. Public result objects expose the converted arrays, but users do not need to call this helper directly.

4 Function

```python
def convert_frequency_axes(
    frequencies_thz: ArrayLike,
) -> tuple[FloatArray, FloatArray, FloatArray]:
    ...
```

The returned arrays are:

1. angular frequency in rad/ps;
2. spectroscopic wavenumber in cm^{-1};
3. quantum energy hf in meV.

5 Mathematical conversions

Angular frequency:

$$\omega = 2\pi f.$$

Wavenumber:

$$\tilde{\nu} = \frac{f}{c}.$$

Because SciPy stores c in m/s and the input is THz,

$$\tilde{\nu}\ [\mathrm{cm}^{-1}] = \frac{f_{\mathrm{THz}} 10^{12}}{100c}.$$

Energy:

$$E = hf,$$

or

$$E\ [\mathrm{meV}] = f_{\mathrm{THz}} \frac{h 10^{15}}{e},$$

where e is the joule value of one electron volt.

6 Inputs and constraints

- the input is one-dimensional and nonempty;
- all values are finite and nonnegative;
- ordering and uniformity are not changed;
- zero is allowed and maps to zero on every axis.

7 Numerical source

The physical constants are read from `scipy.constants`. These conversions are standard dimensional relations, not a borrowed specialized algorithm. SciPy is the direct software source [1].

8 Tests

`tests/test_spectral.py` verifies:

- exact 2π angular conversion;
- the known values 1 THz $\approx$ 33.3564 cm^{-1} and 1 THz $\approx$ 4.13567 meV;
- rejection of multidimensional, negative, empty, and nonfinite inputs.

9 Reference

[1] P. Virtanen, R. Gommers, T. E. Oliphant, et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods* **17**, 261-272 (2020). DOI: 10.1038/s41592-019-0686-2.